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/*****
File Name      : data_validation.sql
Project       : Sales Data Warehouse
Description    : Data Validation Scripts for ETL Validation
Database      : Oracle
*****/

PROMPT =====
PROMPT 1. ROW COUNT VALIDATION
PROMPT =====

SELECT COUNT(*) AS source_count
FROM stg_sales;

SELECT COUNT(*) AS target_count
FROM fact_sales;

PROMPT =====
PROMPT 2. DUPLICATE RECORD VALIDATION
PROMPT =====

SELECT
    order_id,
    COUNT(*) AS duplicate_count
FROM fact_sales
GROUP BY order_id
HAVING COUNT(*) > 1;

PROMPT =====
PROMPT 3. NULL VALUE VALIDATION
PROMPT =====

SELECT *
FROM fact_sales
WHERE customer_id IS NULL
    OR product_id IS NULL
    OR date_id IS NULL
    OR region_id IS NULL
    OR sales_amount IS NULL
    OR quantity IS NULL;

PROMPT =====
PROMPT 4. FOREIGN KEY VALIDATION - CUSTOMER

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PROMPT =====

```
SELECT f.customer_id
FROM fact_sales f
LEFT JOIN dim_customer c
ON f.customer_id = c.customer_id
WHERE c.customer_id IS NULL;
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PROMPT =====

PROMPT 5. FOREIGN KEY VALIDATION - PRODUCT

PROMPT =====

```
SELECT f.product_id
FROM fact_sales f
LEFT JOIN dim_product p
ON f.product_id = p.product_id
WHERE p.product_id IS NULL;
```

PROMPT =====

PROMPT 6. FOREIGN KEY VALIDATION - DATE

PROMPT =====

```
SELECT f.date_id
FROM fact_sales f
LEFT JOIN dim_date d
ON f.date_id = d.date_id
WHERE d.date_id IS NULL;
```

PROMPT =====

PROMPT 7. FOREIGN KEY VALIDATION - REGION

PROMPT =====

```
SELECT f.region_id
FROM fact_sales f
LEFT JOIN dim_region r
ON f.region_id = r.region_id
WHERE r.region_id IS NULL;
```

PROMPT =====

PROMPT 8. DATA RECONCILIATION

PROMPT =====

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SELECT
(
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```

        SELECT SUM(sales_amount)
        FROM stg_sales
    ) AS source_sales,

    (
        SELECT SUM(sales_amount)
        FROM fact_sales
    ) AS target_sales

FROM dual;

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PROMPT =====
PROMPT 9. NEGATIVE SALES VALIDATION
PROMPT =====

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SELECT *
FROM fact_sales
WHERE sales_amount < 0;

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PROMPT =====
PROMPT 10. INVALID QUANTITY VALIDATION
PROMPT =====

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SELECT *
FROM fact_sales
WHERE quantity <= 0;

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PROMPT =====
PROMPT 11. MONTHLY SALES VALIDATION
PROMPT =====

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SELECT
    d.month_name,
    SUM(f.sales_amount) AS revenue
FROM fact_sales f
JOIN dim_date d
ON f.date_id = d.date_id
GROUP BY d.month_name
ORDER BY MIN(d.full_date);

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PROMPT =====
PROMPT 12. REGION-WISE SALES VALIDATION
PROMPT =====

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```
SELECT
    r.region_name,
    SUM(f.sales_amount) AS revenue
FROM fact_sales f
JOIN dim_region r
ON f.region_id = r.region_id
GROUP BY r.region_name;
```

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PROMPT =====
PROMPT 13. PRODUCT CATEGORY VALIDATION
PROMPT =====
```

```
SELECT
    p.category,
    SUM(f.sales_amount) AS revenue
FROM fact_sales f
JOIN dim_product p
ON f.product_id = p.product_id
GROUP BY p.category;
```

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PROMPT =====
PROMPT VALIDATION COMPLETED
PROMPT =====
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